MACHINE LEARNING • LINEAR MODELS

LASSO REGRESSION

Least Absolute Shrinkage and Selection Operator.
A regularized linear model that reduces overfitting and automatically selects features.

01 • LASSO OBJECTIVE

How does Lasso evaluate model performance?

Lasso adds an L1 penalty to the standard OLS loss function.

$$\text{Loss} = \text{SSE} + \lambda \times \Sigma|w_i|$$

The sum of squared errors plus the sum of absolute coefficients.

WHY ABSOLUTE VALUES?

Absolute values create a constant penalty rate, which can drive coefficients completely to zero, unlike squared penalties.

SSE (SUM OF SQUARED ERRORS)

The base prediction error measured against training data.

Λ (LAMBDA)

Regularization strength. High λ means stronger penalty.

$\Sigma|W_i|$ (L1 PENALTY)

Cost based on the absolute size of feature weights.

* The intercept (w_0) is generally NOT penalized.

02 • MECHANISM & FEATURE SELECTION

How L1 regularization alters model weights.

1. SHRINKAGE

Like Ridge, Lasso shrinks coefficient values toward zero to reduce overfitting and variance.

2. EXACT ZEROS

Unlike Ridge, the L1 penalty geometry allows coefficients to become **exactly zero**.

3. FEATURE SELECTION

When a weight becomes zero, that feature is entirely removed from the model. Lasso performs automatic feature selection.

COEFFICIENT IMPACT

Before Lasso (OLS)



All features retain some weight.

After Lasso (λ > 0)



Weak feature (middle) driven exactly to zero.

⚠️ The Lambda Tradeoff

A larger λ applies stronger regularization. If λ is too large, the model may underfit by removing too many useful features.

03 • NUMERICAL EXAMPLE

Calculating the initial Lasso objective function.

INITIAL MODEL STATE

$$\hat{y} = 10 + 100x_1 + 80x_2$$

GIVEN PARAMETERS

$SSE = 500$

$\lambda = 0.1$

$w_1 = 100$

$w_2 = 80$

1. CALCULATE L1 PENALTY

$$Penalty = \lambda \times (|w_1| + |w_2|)$$

$$Penalty = 0.1 \times (|100| + |80|)$$

$$Penalty = 0.1 \times 180$$

$$\mathbf{Penalty = 18}$$

2. TOTAL LASSO OBJECTIVE

$$Loss = SSE + Penalty$$

$$Loss = 500 + 18$$

$$\mathbf{Loss = 518}$$

Note: This evaluates the objective value for the given starting coefficients. The model will learn new, updated coefficients by minimizing this Loss value over time via gradient descent.

Evaluating the strengths and limitations of Lasso.

ADVANTAGES

Reduces Overfitting
Constrains complexity to improve unseen data predictions.

Automatic Feature Selection
Zeroes out useless features entirely.

Creates Sparse Models
Produces simpler models that use memory efficiently.

High-Dimensional Data
Excellent when dataset has more features than samples.

DISADVANTAGES

Risk of Losing Useful Features
If λ is too strong, it drops features that carry predictive signal.

Instability with Multicollinearity
Arbitrarily selects one feature from a correlated group and drops the rest.

Requires Feature Scaling
Unscaled data forces unfair penalties on small-scale features.

Alpha (λ) Tuning Crucial
Requires cross-validation to find the optimal regularization strength.

Understanding when to use which regularization technique.

ASPECT	RIDGE REGRESSION	LASSO REGRESSION
Regularization	L2	L1
Penalty	Sum of squared coefficients	Sum of absolute coefficients
Coefficients	Shrinks toward zero	Can become exactly zero
Feature Selection	No automatic feature removal	Performs feature selection
Main Use	Reducing coefficient magnitude	Reducing coefficients & selecting features

Both Ridge (L2) and Lasso (L1) help control overfitting.

Lasso is specifically preferred when **feature selection** is required to simplify the model.